

No. of Functional Features Included in the Solution

Date	04 July 2026
Team ID	
Project Name	HDI Predictor Using Machine Learning
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	User Input Interface	Accepts HDI-related input values from the user through a web form.	Frontend (HTML/CSS)	Done	High
2	Input Validation	Validates user-entered values before prediction.	Flask Backend	Done	High
3	Data Preprocessing	Processes input data to match the trained model format.	ML Pipeline	Done	High
4	HDI Prediction	Predicts the Human Development Index using the trained Linear Regression model.	ML Model (HDI.pkl)	Done	High
5	HDI Category Classification	Classifies the predicted HDI into Low, Medium, High, or Very High.	Prediction Logic	Done	Medium
6	Result Display	Displays the predicted HDI score and category on the result page.	Flask Templates	Done	High
7	Model Loading	Loads the trained HDI.pkl model during application startup.	Flask Backend	Done	Medium
8	Responsive Web Interface	Provides a simple and user-friendly interface across devices.	Frontend (HTML/CSS)	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2

Metric	Count / Value
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Input Form, Result Page
2	Backend / Logic	3	Input Validation, Model Loading, Prediction Logic
3	Database / Storage	1	HDI.csv Dataset & HDI.pkl Model
4	API / Integration	1	Flask Routing & Request Handling
5	Security / Authentication	0	Not Implemented (Not Required)